

Attack Navigation Layer Coverage Map

Project

SecOpsAI – AI-Powered Behavioral Threat Detection and Adversarially Robust Security Operations Platform

Purpose

The Attack Navigation Layer provides a high-level view of potential attacker movement within the SecOpsAI platform. It identifies critical assets, ATT&CK techniques, affected components, security controls, and detection coverage provided by the platform.

Attack Surface Overview

The following components form the primary attack surface of the SecOpsAI platform:

- API Layer
- Alert Pipeline
- Data Pipeline
- ML Engine
- Dashboards
- Authentication and Authorization Services
- Configuration Files
- Documentation and Reporting Services
- Network Telemetry Sources
- Kafka Message Bus
- PostgreSQL Database

ATT&CK Coverage Map

Entry Point	Target Component	ATT&CK Technique	ATT&CK ID	Potential Impact	Security Controls	Detection Coverage
Public API	API Services	Exploit Public-Facing Application	T1190	Unauthorized Access	Authentication, Authorization, Input Validation	Rules
Public API	Authentication Services	Brute Force	T1110	Account Compromise	MFA, Rate Limiting	Rules + ML

Entry Point	Target Component	ATT&CK Technique	ATT&CK ID	Potential Impact	Security Controls	Detection Coverage
Credentials	Authentication Services	Valid Accounts	T1078	Account Takeover	JWT Security, Access Control	Rules + ML
External Network	Infrastructure	Network Service Discovery	T1046	Reconnaissance	Zeek, Suricata Monitoring	Rules + ML
External Network	Network Services	Application Layer Protocol	T1071	Command and Control Activity	Traffic Analysis	Rules + ML
DNS Traffic	Data Pipeline	DNS Protocol	T1071.004	DNS Tunneling	DNS Monitoring	Rules + ML
Internal Host	Infrastructure	Remote Services	T1021	Lateral Movement	Authentication Controls	Rules + ML
Internal Host	Data Pipeline	Exfiltration Over C2 Channel	T1041	Data Exfiltration	Behavioral Analytics	ML
Internal Host	Infrastructure	Data Obfuscation	T1001	Evasion of Detection	Behavioral Analytics	ML
User Dashboard	Dashboard Services	Abuse Elevation Control Mechanism	T1548	Privilege Escalation	RBAC	Rules + ML
Internal Host	Infrastructure	Masquerading	T1036	Process Impersonation	Behavioral Analytics	ML
Data Sources	ML Engine	Training Data Poisoning	Custom AI Threat	Model Corruption	Dataset Validation	ML
External User	ML Engine	Model Evasion	Custom AI Threat	Detection Bypass	Adversarial Testing	ML
External User	ML Engine	Model Inversion	Custom AI Threat	Sensitive Data Exposure	Rate Limiting	ML
External User	ML Engine	Membership Inference	Custom AI Threat	Training Data Leakage	Dataset Monitoring	ML

Attack Flow Examples

Scenario 1: Unauthorized API Access

1. Attacker targets a public API endpoint.
2. Attempts credential guessing (T1110).
3. Obtains valid credentials (T1078).
4. Gains unauthorized access to protected resources.
5. Authentication controls and anomaly detection generate alerts.

Scenario 2: DNS Tunneling and Data Exfiltration

1. Attacker compromises an internal host.
 2. Establishes covert communication using DNS (T1071.004).
 3. Exfiltrates sensitive information (T1041).
 4. Behavioral analytics identify abnormal traffic patterns.
 5. Security alerts are generated for investigation.
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Scenario 3: Lateral Movement and Privilege Escalation

1. Attacker compromises a low-privilege account.
 2. Uses remote services for lateral movement (T1021).
 3. Attempts privilege escalation (T1548).
 4. RBAC controls block unauthorized access.
 5. Monitoring systems generate alerts and log activity.
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Scenario 4: ML Data Poisoning Attack

1. Malicious data enters the ingestion pipeline.
 2. Poisoned data reaches the ML engine.
 3. Model accuracy degrades.
 4. Dataset validation identifies anomalies.
 5. Suspicious datasets are quarantined for review.
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Security Monitoring Coverage

The platform provides monitoring and detection for:

- Valid Account Abuse (T1078)
- Brute Force Attacks (T1110)
- Exploitation of Public-Facing Applications (T1190)
- Network Service Discovery (T1046)
- Command and Control Activity (T1071)
- DNS Tunneling (T1071.004)
- Remote Services / Lateral Movement (T1021)
- Exfiltration Over C2 Channels (T1041)
- Data Obfuscation (T1001)
- Privilege Escalation (T1548)
- Disable or Modify Tools (T1562)

- Masquerading (T1036)
 - ML Model Evasion
 - Data Poisoning
 - Model Inversion
 - Membership Inference
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Conclusion

The Attack Navigation Layer maps attacker movement throughout the SecOpsAI environment to MITRE ATT&CK techniques and platform detection capabilities. This coverage map supports threat modeling, ATT&CK Navigator development, detection engineering, adversarial testing, and continuous security improvement activities.